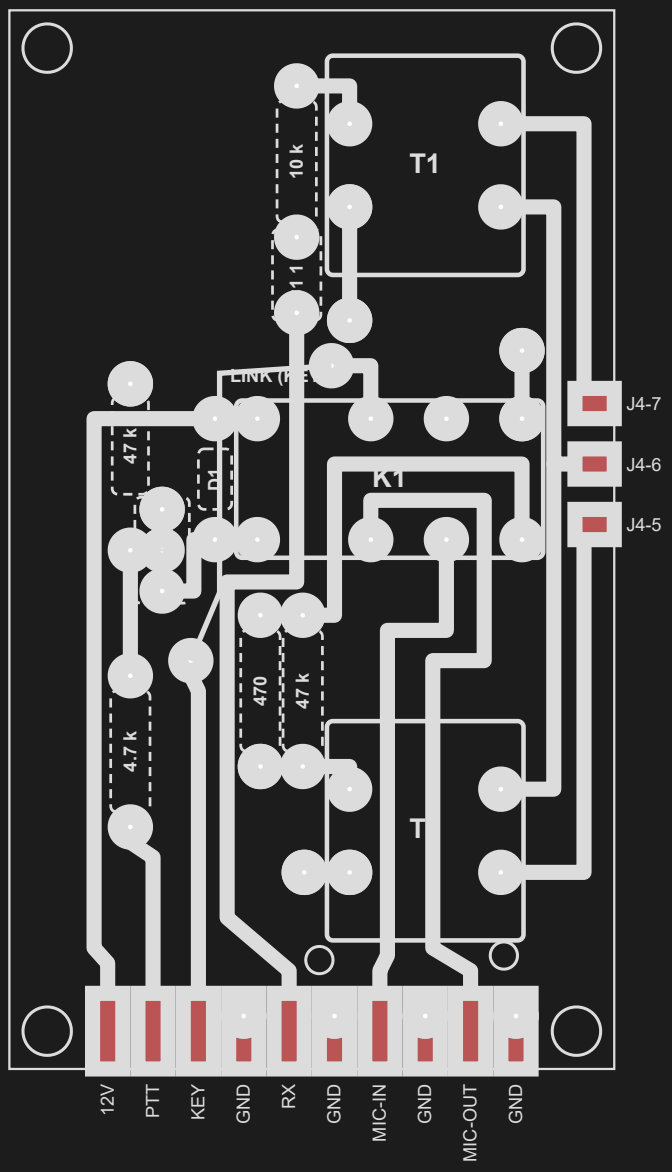
PTT / data audio board — parts and tracks

40 × 70 mm, 2 × size, seen from the parts side with the tracks shown through the board. Top = LCD end, left = front of the radio.

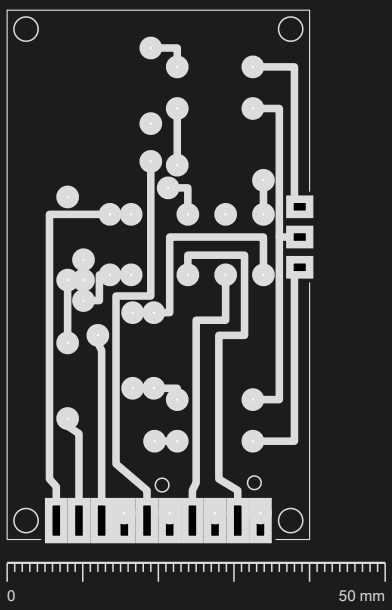


- **Single-sided board, as built.** Tracks and edge pads on the underside (red); no top copper.
- **Green rings = GND**, joined by insulated wires on the parts side and short jumpers underneath: GND pad next to KEY, 47 k and BC547 emitter; last GND pad, T1 and relay NO; GND pad next to RX, T2 and 470 Ω. Check with the meter that all GND points are joined.
- **LINK** (blue) = one insulated wire on the parts side, around the relay's coil end, taking KEY across the other tracks.
- Relay: coil end on the left. Upper row = KEY pole, lower row = audio pole. D1 band at the top (12 V).
- BC547, pins in a column: E top, B middle, C bottom, flat side towards the left edge. Check E-B-C on your part. Check each transformer pair with the meter: each side must be one winding.

Pad	Goes to	On the board
12V	radio +12 V (red)	relay coil + D1 band
PTT	VariBeam D9 (yellow)	4.7 k to the BC547 base; 47 k base to GND
KEY	KEY line (white). **Never the T line**	LINK, relay upper row COM; NO = GND
RX + GND	PC-300 pin 22, coax	1 µF, 10 k, T1
MIC-IN + GND	mic socket, coax	relay lower row NC
MIC-OUT + GND	PC-200 pin 17, coax	relay lower row COM
J4-7 / J4-6 / J4-5	J4 pins 7, 6, 5, twisted	T1 and T2 PC side; J4-6 floating, never to GND

Etching artwork — underside copper

Print at 100 % / Actual size; the ruler must read 50 mm. Already correct for toner transfer: do not mirror it again.



- Tracks 1.0 mm, pads ø 2 mm, clearance at least 0.6 mm (checked by the script).
- Drill the small holes 0.9–1.2 mm, screws 3.2 mm, cable tie 1.8 mm.
- The outline is an alignment aid, with gaps at the edge pads. Remove any copper left on it before soldering.
- Before etching: stand the real relay, transformers, capacitor and BC547 on this print and check the pins.

